

TECHNICAL & ECONOMIC WHITEPAPER

Causeway Causeway Technical & Economic Whitepaper

An AI trading-intelligence and verifiable reasoning layer for prediction markets: from Polymarket data, risk previews, and Arc proofs toward x402 agent services, parallel market worlds, and user-governed delegated execution.

VERSION

v0.7

DATE

2026-05

STATUS

Detailed Draft

SCOPE

Market + Arc + Swarm

This document describes Causeway's market thesis, product scope, technical architecture, Arc integration, future x402 module, swarm prediction engine, economic model, risk boundaries, and roadmap. It is not investment advice, legal advice, brokerage material, a return promise, or an offer of automated trading. Any real transaction should remain user-directed and user-confirmed.

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01 / EXECUTIVE SUMMARY

Executive Summary

Causeway is an AI trading-intelligence and verifiable reasoning layer for prediction markets. It organizes real Polymarket market data, AI reasoning, risk previews, order previews, Arc proof, and Signal Track Record into an auditable workflow.

Version 0.7 upgrades the long-term vision from single-model judgment to Causeway’s own Swarm Prediction Engine: a system that builds parallel market worlds, coordinates role-specialized agents, simulates event propagation, and turns the output into tradable, verifiable, reviewable market-intelligence objects.

02 / MARKET CONTEXT: PREDICTION MARKETS AS PROBABILITY INFRASTRUCTURE

Market Context: Prediction Markets as Probability Infrastructure

Prediction markets are evolving from event-betting interfaces into probability data layers for politics, macro events, sports, corporate events, and on-chain activity. Prices are not only trades; they are public signals that aggregate dispersed information in real time.

As markets scale, users no longer merely need discovery. They need help deciding which prices already reflect information, which related markets lag, and which signals are only liquidity noise.

03 / PROBLEM: WHY PREDICTION MARKETS STILL LACK AN INTELLIGENCE LAYER

Problem: Why Prediction Markets Still Lack an Intelligence Layer

Markets are networks, but most interfaces still present lists. AI can explain text, but often lacks verifiable input snapshots, reasoning traces, order previews, and post-event performance records.

Causeway addresses the full chain from event to market network, from probability to edge, from reasoning to user confirmation, and from pre-event record to post-event audit.

04 / ACADEMIC FOUNDATION AND VALUE CALCULATION FRAMEWORK

Academic Foundation and Value Calculation Framework

Prediction-market research shows that binary contract prices can approximate event probabilities under ideal conditions, but real markets contain spreads, fees, slippage, limits, source risk, and resolution ambiguity. Causeway therefore separates research probability from executable probability.

Value does not come from “AI thinks it is more likely.” It comes from positive edgeNet after transaction friction, source uncertainty, rule risk, and portfolio correlation are deducted.

```
p_mid = (bestBid + bestAsk) / 2
p_exec_yes = ask_yes, p_exec_no = ask_no
q_ai = calibratedForecast(event | marketSnapshot, sourceObjects, reasoningTrace)
rawEdge_mid = q_ai - p_mid
Separate midpoint probability from executable price to avoid overstating edge.
```

```
EV_token_yes = q_ai * 1 + (1 - q_ai) * 0 - ask_yes - cost_per_token
ROI_yes = EV_token_yes / ask_yes
edgeNet = q_ai - ask_yes - feeRate - slippageBps - ruleRiskHaircut - sourceRiskHaircut
BUY only if edgeNet > minEdge, depthAtLimit > targetSize, timeToClose > minWindow
Only edgeNet that remains positive after costs, slippage, rule risk, and source risk should enter trade preview.
```

```
q_adj = clamp(0.5 + confidence * (q_ai - 0.5), 0.01, 0.99)
b = (1 - p_exec) / p_exec
kellyFull = (b * q_adj - (1 - q_adj)) / b = (q_adj - p_exec) / (1 - p_exec)
sizeUsd = bankroll * min(max(0, lambda * kellyFull), capMarket, capPortfolio, capCorrelation)
Conservative Kelly shrinks model probability toward 50% and caps sizing by capacity, correlation, and budget.
```

```
Underround:  $\sum \text{ask}_i + \text{fees} + \text{slippage} < 1$ 
profitFloor_buyBasket =  $1 - \sum \text{ask}_i - \text{fees} - \text{slippage} - \text{settlementRisk}$ 
If event B implies event A, then  $P(B) \leq P(A)$ 
semanticEdge = violation * relationConfidence * min(liquidityScore_A, liquidityScore_B)
Mutually exhaustive baskets and semantic implication help detect inconsistency, but execution depth and rule compatibility must be verified.
```

```
Brier_mean = mean((q_ai - y)^2)
Logloss_mean = mean(-[y * ln(q_ai + eps) + (1 - y) * ln(1 - q_ai + eps)])
CalibrationError =  $\sum_k n_k / N * |\text{mean}(q_{ai} \text{ in bucket } k) - \text{mean}(y \text{ in bucket } k)|$ 
signalScore = z(edgeNet) + z(confidence) + z(liquidity) - z(spreadRisk) - z(sourceRisk) - z(correlationRisk)
Long-term evaluation must include calibration, log loss, Brier score, and non-executed opportunity performance, not only PnL.
```

05 / PRODUCT DEFINITION

Product Definition

Causeway is a prediction-market Trader Intelligence Layer. Its core output is not a chat answer but a market intelligence object: root market, candidate markets, semantic relations, AI fair odds, market odds, edgeNet, risk explanations, order preview, Arc proof, and track record.

The default boundary is user governance: AI expands the thesis and previews risk; the user keeps final confirmation and control of funds.

06 / WHAT WE HAVE SOLVED

What We Have Solved

Causeway separates market titles from real tradable outcome tokens and models events, markets, outcomes, CLOB token IDs, prices, books, and liquidity.

Arc proof can anchor reasoning-trace hashes on Arc; Arc USDC premium can verify membership through payment intents and Transfer logs.

07 / ARCHITECTURE AND DATA MODEL

Architecture and Data Model

The architecture combines React, NestJS, Prisma/PostgreSQL, Polymarket Gamma/CLOB data, structured AI outputs, Arc proof, and USDC payment verification.

Core objects include Market Object, Inference Object, Causal Script, Order Intent, Arc Proof Capsule, and Signal Track Record.

08 / AI TRADER INTELLIGENCE: FROM PROBABILITY TO ACTION PREVIEW

AI Trader Intelligence: From Probability to Action Preview

A mature system should not only output BUY. It must be able to return WATCH, VERIFY FIRST, or AVOID. No Trade is a capability because it shows awareness of liquidity, sources, rules, and portfolio risk.

Position sizing comes from edgeNet, confidence, order-book depth, spread, budget, correlation, and conservative Kelly, not from a fixed amount.

09 / ARC PROOF: VERIFIABLE AI REASONING RECORDS

Arc Proof: Verifiable AI Reasoning Records

Prediction markets verify today's judgment against future outcomes. AI signals must therefore prove they existed before resolution. Arc proof anchors the reasoning-trace hash on-chain, reducing silent rewriting risk.

Arc does not replace the Polymarket trading path. It is a low-cost, stablecoin-native, EVM-compatible audit layer.

10 / ARC USDC PREMIUM: MEMBERSHIP, VERIFIABLE PAYMENT, AND AGENT ECONOMY

Arc USDC Premium: Membership, Verifiable Payment, and Agent Economy

Current membership uses Arc USDC payment intents to unlock premium signals, full reasoning traces, Arc proof, and higher analysis limits.

In the future, Causeway may use x402 calls settled on Arc to unify subscriptions, per-report unlocks, APIs, data sources, and agent capabilities into a finer-grained payment framework.

x402 Agent Service Layer: Future Agent-Service Protocol Layer

x402 is not the trading-execution protocol in Causeway. It is an agent-to-service payment and access protocol for paid data, verification services, specialized agents, reports, and APIs.

Causeway orchestrates market graphs, permission governance, risk controls, and performance tracking; x402 handles paid access; Arc records proof, payment record, and reputation.

Swarm Prediction Engine: From Parallel Market Worlds to Predicting Anything

Causeway’ s long-term goal is not to make a single AI judge one market once. It is to build a swarm prediction engine native to prediction markets. Starting from a real event, the system constructs parallel market worlds and lets role-specialized agents debate evidence, counterexamples, probability, risk, and execution boundaries.

A parallel market world is not generic storytelling. It is prediction-market-native simulation: each world must explain how an event travels through the market graph, which odds move, which edge appears, which risk activates, and what verifiable record remains.

The engine compresses agent debate into structured market-intelligence objects: scenario tree, affected markets, probability shift, AI fair odds, edgeNet, risk flags, suggested size, Arc proof hash, and track record entry.

Agent Society: Role-Specialized Agents

Agent	Role	Output
Research Agent	Collects events, market context, history, and candidate sources.	sourceObjects, event summary, market candidates
Market Graph Agent	Detects related markets, semantic implication, mutual exclusion, and exposure overlap.	market graph, relation type, impact direction
Probability Agent	Turns evidence and scenarios into calibrated probabilities.	AI fair odds, probability shift, confidence
Skeptic Agent	Searches for counterexamples, rule ambiguity, weak sources, and overreach.	counterarguments, changeMyMind, risk flags
Verification Agent	Traces claims to primary or authoritative sources.	verification status, source confidence, conflict report
Risk Agent	Evaluates liquidity, spread, slippage, correlation, and position caps.	edgeNet, risk budget, position cap
Execution Guard	Decides whether a signal may enter preview or delegated execution.	BUY / WATCH / AVOID, order gate, emergency stop
Report Agent	Turns debate and disagreement into an auditable prediction report.	prediction report, scenario tree, audit summary

```
scenarioValue_s = Σ_i edgeNet_i,s * tradability_i,s * confidence_s - portfolioRisk_s
swarmConsensus = weightedMedian(q_agent_1, ..., q_agent_n; weights = reputation * calibration)
disagreementRisk = variance(q_agent_1 ... q_agent_n) + sourceConflict + ruleAmbiguity
finalAction = gate(swarmConsensus, edgeNet, disagreementRisk, liquidity, userPolicy)
```

Swarm intelligence is not simple voting; it combines reputation, calibration, disagreement, source conflict, and user policy into an action gate.

13 / USER WORKFLOW AND PRODUCT EXPERIENCE

User Workflow and Product Experience

The user connects a wallet, selects a root outcome, reviews the AI reasoning graph and causal script, enters order preview, confirms dry-run or real CLOB signing, and can anchor the trace on Arc.

14 / RISK, GOVERNANCE, AND COMPLIANCE BOUNDARY

Risk, Governance, and Compliance Boundary

The system does not custody private keys, promise returns, or bypass confirmation. Automation requires explicit authorization, budget limits, scope limits, emergency stops, and revocation.

15 / FUTURE PROBLEMS

Future Problems

Next stages must solve source authenticity, multi-agent calibration, provable Signal Track Record, x402 service-market quality, and governance for bounded delegated execution.

16 / FIVE-PHASE ROADMAP

Five-Phase Roadmap

The roadmap starts with Polymarket market data foundation, then expands to reasoning model, real-time scenario generation, verification and Arc proof, and optional delegated AI execution.

Roadmap

Phase	Theme	Description
Phase 01	Market Data Foundation	Understand event, market, outcome, token ID, price, liquidity, order book, and rules before expanding external sources.
Phase 02	Reasoning Model	Map one event to related markets with relevance, direction, confidence, tradability, recommendation, and disagreement checks.
Phase 03	Real-Time Scenario Generation	Convert event streams into market response scripts and gradually expand toward parallel market-world simulation.
Phase 04	Verification & Arc Proof	Verify bottom-layer sources before action and anchor reasoning traces on Arc for auditability.
Phase 05	Delegated AI Execution	Allow optional, bounded delegation only after permissions, budgets, revocation, and audit controls mature.

17 / ECONOMIC MODEL AND VALUE CAPTURE

Economic Model and Value Capture

Revenue includes Arc USDC premium, builder attribution, Signal API, x402 Agent Service Layer, Swarm Prediction Reports, Team Workspace, and Agent Marketplace.

Revenue Model

Model	Description	Stage
Premium Subscription	Arc USDC membership for deeper models, complete traces, Arc proof, monitoring; future x402-based usage packs.	Phase 1-3
Builder Attribution	User-confirmed Polymarket orders can be attributed through builder code.	Phase 1-5
Signal API	Structured signals, market graphs, risk reports, and track record APIs.	Phase 2-4
x402 Agent Service Layer	Paid access to data sources, verification services, specialized agents, and reports.	Phase 3-5
Swarm Prediction Reports	Professional prediction reports generated from parallel market worlds and agent debate.	Phase 3-5
Agent Marketplace	Specialized agents and verification modules settle through Arc USDC and x402.	Phase 4-5

18 / MOAT, METRICS, AND CONCLUSION

Moat, Metrics, and Conclusion

Causeway’ s moat comes from market-structure understanding, reasoning data loops, Arc-verifiable history, x402 agent-service access, swarm prediction capability, and user-governed boundaries.

The endgame is predicting anything while preserving user governance: the user enters an event; the system understands markets, organizes agents, builds parallel market worlds, verifies facts, generates signals, preserves proof, and lets the user decide whether to act.

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